NEO SEDAI

Core Paradigms: Supervised, Unsupervised, and Reinforcement Learning

1. Supervised Learning

In Supervised Learning, the algorithm learns from a labeled dataset where both input features (X) and target labels (Y) are provided. The goal is to learn a mapping function f(X) → Y to predict outcomes for unseen data.

- **Regression:** Predicts continuous numerical values (e.g., house prices, salary estimation). Algorithms: Linear Regression, Decision Trees.
- **Classification:** Predicts discrete categorical labels (e.g., spam vs. non-spam, disease detection). Algorithms: Logistic Regression, Support Vector Machines (SVM), Random Forests.

2. Unsupervised Learning

Unsupervised Learning deals with unlabeled data (X only). The algorithm independently discovers hidden patterns, underlying structures, or groupings within the dataset without explicit human guidance.

- **Clustering:** Groups similar data points together based on feature similarity (e.g., customer segmentation). Algorithms: K-Means Clustering, Hierarchical Clustering, DBSCAN.
- **Dimensionality Reduction:** Compresses high-dimensional feature spaces while preserving essential variation. Algorithms: Principal Component Analysis (PCA), t-SNE.
- **Association Rule Learning:** Discovers relationships between variables in large databases (e.g., market basket analysis).

3. Reinforcement Learning (RL)

Reinforcement Learning is a feedback-driven learning paradigm where an **Agent** interacts with a dynamic **Environment** to achieve a goal through trial and error.

- **Mechanism:** The agent takes an *Action*, transitions to a new *State*, and receives a positive *Reward* or negative *Penalty*.
- **Objective:** Learn an optimal *Policy* to maximize the total cumulative reward over time.
- **Applications:** Game playing (AlphaGo, chess bots), autonomous vehicle navigation, robotics control, and algorithmic trading.

> **Key takeaway for GitHub log:** Supervised = Learning with a Teacher (Labeled Data); Unsupervised = Self-Discovery of Patterns (Unlabeled Data); Reinforcement = Trial and Error via Rewards and Penalties.